Kabadiwala Connect — Project Record

Smart India Hackathon 2026 · Problem statement SIH26229 · Ministry of Mines (Clean & Green Technology)

This document records everything the team built, step by step, in simple language: what the app does, how it works inside, the tools we used, the problems we hit and how we solved them. It is updated after every build phase.

Last updated: 24 October 2026, night (core app + AI recogniser + real data, tested on a real phone)

1. The problem in one page

India produces a huge amount of electronic waste (old phones, TVs, wires, batteries, circuit boards). Most of it is collected by informal scrap collectors called **kabadiwalas**. They usually:

- sell to the nearest local dealer at a low price, because they don't know the fair rate;
- have no record of what they sold, to whom, or how much they were paid;
- sometimes break or burn the waste (for example burning cables for copper), which is dangerous for their health and the environment;
- often cannot read well, use cheap Android phones, and have weak or no internet.

Meanwhile, **authorised recyclers** (registered with the government to recycle e-waste safely) want more material but can't reach these collectors.

2. Our solution

Kabadiwala Connect is an Android app that works **without internet** and **without needing to read**. A collector can:

1. Take a photo of the scrap.
2. Pick what it is by tapping a picture (circuit board, copper wire, battery...).
3. Enter the weight with big + and − buttons.
4. **Hear** a fair price spoken aloud in their own language.
5. See nearby **authorised** recyclers, ranked by best rate, distance and pickup.
6. Hand over the scrap and get a **QR code** — a tamper-proof digital receipt.
7. The recycler scans the QR, checks it, and records the payment (cash or UPI).
8. The collector's **earnings ledger** updates. Everything syncs when internet comes back.

Who uses it

User	What they do in the app
Collector (kabadiwala)	Sells scrap, hears prices, finds recyclers, keeps an earnings record
Recycler	Scans the collector's QR, confirms weight and payment, sets their rates
Admin (our team)	Updates the price list and recycler list

Design rules we follow (for low-literacy users)

Rule	Why
No screen needs reading: every button = big icon + 1–2 words + sound	Many users can't read comfortably
At most 4 main choices per screen; buttons at least 56 dp tall	Easy to tap, not confusing
Prices are always spoken, never only shown	Numbers on screen are not enough
Colour is never the only signal: green + tick = done, amber + clock = waiting, red + warning = danger	Works for colour-blind users too

Rule	Why
Works offline after first login	Internet is patchy where collectors work
Cash is as important as UPI	Most collectors are paid in cash
All text comes from language files, never typed into the code	So we can add languages easily

3. Technology we chose (and why)

Part	Tool	Why we chose it (simple reason)
App	**Flutter** (Dart language)	One codebase, fast on cheap Android phones
State management	**Riverpod**	Keeps screens updated automatically when data changes
Database	**Cloud Firestore** (Firebase)	Works offline: saves on the phone first, uploads later
Login	**Firebase Phone Auth**	Collectors log in with their phone number + OTP, no passwords
Voice	**Android Text-to-Speech** + recorded clips	Speaks prices; recorded clips for languages Android can't speak
Tamper-proof receipt	**SHA-256 hash** + **QR code**	Any change to the record changes the code, so cheating is visible
Scripts	**Python** (pandas, firebase-admin)	Turns the team's spreadsheet data into app data
Cost	Firebase free "Spark" plan	No money needed for the prototype

Minimum phone: Android 7 (released 2016) or newer. The app works on all newer versions up to Android 16.

4. How we set up the computer (Mac)

Before writing any code, we installed the tools. Here is what each one is for:

Tool	What it does
Homebrew	An "app store" for developer tools on Mac
Java 17	Needed by Android's build system
Flutter 3.47	The framework the app is written in (includes Dart)
Android SDK 36 + build tools	Turns our code into an Android app (APK)
Android NDK	Compiles small native parts used by some plugins
Gradle 9.3	Android's build tool; downloads libraries and builds the APK
Firebase CLI	Lets us log in to Firebase and upload security rules from the terminal
FlutterFire CLI	Connects the Flutter app to our Firebase project
Python virtual environment	A private box for our Python scripts' libraries
Xcode Command Line Tools	Apple's compiler; `flutter test` needs it

The full step-by-step guide for teammates is in `docs/SETUP.md`.

Problems we hit during setup, and how we fixed them

Problem	Cause	Fix
Downloads were very slow (about 200 KB/s)	Slow internet connection	Ran downloads in the background and resumed them; switched to smaller downloads where possible
First Android build failed: "Operation timed out" while downloading Gradle	Our internet provider could not reach **one** of GitHub's 4 download servers (185.199.109.133). Java kept picking that one.	Downloaded Gradle with `curl` (which picked a working server), checked its SHA-256 checksum, and placed it where Gradle expects it
`flutter test` failed: "CarbonCore/Folders.h file not found"	The Mac's Command Line Tools installation was damaged (a file was missing)	Deleted and reinstalled the Command Line Tools
Android NDK download (900 MB) kept stalling	Slow internet; Android's downloader deletes the partial file if stopped	Downloaded it with `curl` in resume mode, verified the official SHA-1 checksum, then installed it manually
Wanted minimum Android 6 (API 23)	Current Flutter no longer supports Android 6	Set the minimum to Android 7 (API 24); almost no phones still run Android 6
Wikimedia stopped our photo downloads ("too many requests")	We downloaded too fast	Used Wikimedia's standard image size and waited between downloads

Firebase setup (done in the browser by the team)

1. Created the Firebase project `kabadiwala-connect` (free Spark plan, no Google Analytics).
2. Created the Firestore database `(default)`, **Standard edition**, location **Mumbai (asia-south1)** — closest to our users, so faster.
3. Turned on **Phone sign-in** and added test phone numbers (fake numbers with a fixed OTP, so we can test without real SMS).
4. Logged in with `firebase login`.
5. Downloaded a **service-account key** (a secret file that lets our Python script write data). It is stored as `scripts/serviceAccount.json` and is **never** uploaded to git.
6. Ran `flutterfire configure`, which registered the Android app with Firebase and generated the connection settings (`firebase_options.dart`, `google-services.json`).

5. How the project is organised

Folder	What's inside
`app/`	The Flutter Android app

Folder	What's inside
`app/lib/core/`	Shared building blocks: languages, audio, theme, hashing, location, sync
`app/lib/data/`	Data models, the code that reads/writes the database, offline seed data
`app/lib/features/`	One folder per screen group: onboarding, home, selling, prices, recyclers, handover, ledger, safety, recycler mode
`app/assets/`	Language files, sounds, icons, offline seed data
`app/test/`	Automatic tests that check the logic is correct
`data/`	Spreadsheets (CSV) the team fills from the field study: materials, prices, recyclers, areas
`scripts/`	Python scripts that load the spreadsheets into the app and the database
`docs/`	Plan, setup guide, this record
`firestore.rules`	Security rules: who can read and write what in the database

6. What we built, phase by phase

The plan has 14 phases (0–13). Each phase ends with automatic tests, a code check (`flutter analyze`) and a saved version in git.

Phase 0 — Setup and scaffolding ✓

Goal: an empty app that runs, connects to Firebase, and has the right look.

What we made:

- Created the Flutter app (Android only, package `in.kabadiwalaconnect`).
- Added all the libraries the plan needs (camera, location, voice, QR, database...).
- Created the folder layout above.
- **Big-button theme:** every button at least 56 dp tall, large bold text, high-contrast colours, and three status colours (green, amber, red) that are always paired with an icon.
- A placeholder home screen that shows a **green tick** if Firebase connected or an **amber clock** if not. The app never crashes or blocks if Firebase is missing — it just keeps working.

Phase 1 — Data layer and seed data ✓

Goal: the app knows about materials, prices and recyclers, even with no internet on the very first run.

The data (spreadsheets in `data/`):

File	What it holds
`materials.csv`	8 scrap types: circuit board, copper wire, phone battery, vehicle battery, old TV, motor, mixed plastic, old phone. Each has an icon and a hazard level (none / low / high)
`prices.csv`	Price range per kg (low–high), last week's price, and the local dealer's price (for comparing earnings later)
`recyclers.csv`	Recycler name, phone, location, whether authorised, registration number, pickup or not, and their rate per material

File	What it holds
`areas.csv`	A few areas with map coordinates, used if a collector doesn't allow location access

All rows are clearly marked **placeholder** for now. We will replace them with real data from our field study. We never invent real business names or phone numbers.

How the data gets into the app:

- `scripts/csv_to_assets.py` turns the spreadsheets into small files that are **built into the app** (`app/assets/seed/`). This is the offline backup.
- `scripts/seed_firestore.py` uploads the same data to the Firestore database.

Money and weight are stored as whole numbers: money in **paise** (Rs 1 = 100 paise) and weight in **grams**. Computers make small rounding errors with decimal numbers (for example 0.1 + 0.2 = 0.30000000000000004), which is unacceptable for money. Whole numbers never have this problem.

Offline-first reading: when a screen needs the price list, the app shows the built-in copy **immediately**, then switches to the live database copy as soon as it's available. So the app is never empty and never waits for internet.

Offline-first writing: when the collector saves something (a lot, a handover), Firestore saves it **on the phone instantly** and uploads it when internet returns. Every record gets a unique ID made on the phone (a UUID), so even if the same record is sent twice, it never becomes two records.

Data models (the "shape" of each record): Material, Price, Recycler, Area, User, Lot (one batch of scrap), Handover, Confirmation, Ledger entry.

One change from the original plan: the plan had the recycler update the collector's handover record. But if **both** phones are offline, the recycler can't update a record that hasn't reached the server yet. So the recycler writes a separate **Confirmation** record with the same ID. The collector's app joins them. This works even when both phones are offline.

Security rules (`firestore.rules`, uploaded to Firebase):

Data	Who can read	Who can write
Materials, prices, areas	Anyone logged in	Admin only
Recyclers	Anyone logged in	Admin; an invited recycler can link their login and edit only their own rates
A user's profile	Only that user	Only that user (cannot make themselves admin)
Lots, ledger	Only the collector who owns them	Only that collector
Handovers	The collector and the receiving recycler	Collector creates it; after that it can **never be changed**
Confirmations	The collector and the recycler	Only the recycler who received the scrap

Phase 2 — Languages and voice ✓

Goal: the app speaks and shows text in the user's language, and switching language is instant.

Languages (12):

Code	Language	How it's spoken	Why it's included
hi	Hindi (default)	Android text-to-speech	Most collectors understand it
gbm	Garhwali	Recorded clips (Android can't speak it); Hindi voice until recorded	Local language of Uttarakhand
mr	Marathi	Android text-to-speech	Required by the problem statement
en	English	Android text-to-speech	For judges and recyclers
bn	Bengali	Android text-to-speech	Many collectors in North Indian cities come from West Bengal
bho	Bhojpuri	Hindi voice until clips are recorded	Many collectors come from Bihar and East UP
pa	Punjabi	Android text-to-speech	Spoken around Roorkee / West UP / Punjab
ur	Urdu	Android text-to-speech; the whole screen flips right-to-left	Widely spoken in West UP
gu	Gujarati	Android text-to-speech	Large e-waste hubs in Gujarat
ta	Tamil	Android text-to-speech	Large e-waste hubs in Tamil Nadu
te	Telugu	Android text-to-speech	Large e-waste hubs in Telangana / Andhra
kn	Kannada	Android text-to-speech	Bengaluru is a major e-waste city

Honesty note: the developers wrote the translations for every language except English. Each file is marked draft — check with a native speaker, and that check must happen before field use. Garhwali is still a copy of the Hindi text until the team translates it.

Choosing a language without reading: the first screen shows only **4 big choices** (Hindi, Garhwali, Marathi + "More languages"), keeping to the 4-choice rule. "More" shows all 12. Each tile says its language's name out loud on the first tap and chooses it on the second. The language can be changed any time with the language (translate) button at the top of the home screen.

Each language has one file, for example `assets/i18n/hi.json`. Text can have slots, e.g. "You will get about {amount} rupees" — the app fills in the number.

How the app decides how to speak (3 steps):

1. If a **recorded clip** exists for this sentence in this language, play it. Numbers are built from recorded number clips.
2. Otherwise, if the phone's **text-to-speech** knows the language, speak it.
3. Otherwise, speak it in **Hindi**.

Speaking numbers from recorded clips (NumberSpeaker): Hindi-family numbers are irregular (e.g. 21 = "ikkees", 45 = "paintalees"), so we record every number from 0 to 99 plus "hundred", "thousand", "lakh", "rupees", "about". The app breaks any amount into these pieces using the Indian system:

Amount	Clips played
Rs 340	about · 3 · hundred · 40 · rupees
Rs 1,340	about · 1 · thousand · 3 · hundred · 40 · rupees

Amount	Clips played
Rs 1,50,000	about · 1 · lakh · 50 · thousand · rupees

Hidden audio-check screen: long-press the app title to open a list of every sentence with a play button, so the team can check each language.

`scripts/check_i18n.py`**:** checks that every language file has every sentence, and lists which recorded clips are still missing.

Phase 3 — Onboarding and home ✓

Goal: a first-time user can pick a language, log in, set a PIN and reach the home screen; after that the app opens **offline with just the PIN**.

Screen	What happens
S1 Language	Big buttons with each language's name in its own script. **Tap once to hear it, tap again to choose it** — so nobody has to read
S2 Login	Phone number on a big number pad (+91 is fixed), then the 6-digit SMS code. For testing we use Firebase test numbers (no real SMS)
S3 PIN	Make a 4-digit PIN (typed twice). Every time the app starts, it asks for the PIN — this works **without internet**
S4 Role	Only shown if this phone number was invited by a recycler. Everyone else goes straight in as a collector
S5 Home	Four big picture tiles: Sell scrap, Today's rates, My earnings, Safety

How it works inside:

- **The app remembers the login.** Firebase keeps the signed-in user on the phone, so after the first online login the app never needs internet to open.
- **The PIN is never stored.** Only a SHA-256 fingerprint of it (mixed with the user ID) is saved. A 4-digit PIN is a convenience lock for the phone, not strong security, and we say so.
- **Every screen talks.** A shared screen frame speaks the screen's instruction when it opens, and has a speaker button to repeat it, plus a small **sync icon**: green cloud = everything sent, amber cloud = waiting to send, grey cloud = no internet (the app still works). Tapping the sync icon says what it means.
- **Every big tile** speaks its name when tapped. Long-press only speaks (for "what is this?").
- **Recycler invitation:** a recycler listing in the database has an `ownerPhone`. When that phone number logs in, it may choose recycler mode, and its login is linked to the listing. The security rules check the phone number on the server, so nobody can pretend to be a recycler.
- To make phone login work on Android, we registered the app's signing fingerprints (SHA-1 and SHA-256) with Firebase from the terminal.

Phase 4 — Selling scrap (photo → category → weight → spoken price) ✓

Goal: the heart of the demo. In airplane mode, a collector can photograph scrap, pick what it is, set the weight and **hear a fair price**.

Screen	What the collector does
S6 Photo	Big round camera button, up to 3 photos; a red X on a photo removes it to retake
S7 Category	Picture tiles in 2 columns. Dangerous items (batteries, old TVs) have a **red border and warning icon** and say a warning out loud
S8 Weight	A huge number with −1, −½, +½, +1, +5 kg buttons. Every change is spoken ("2.5 kilo")
S9 Price	Says **"You will get about Rs 300"** and shows the range (Rs 270–Rs 330). The lot is saved immediately, even offline

The price formula (from the plan, all in whole paise):

- low = weight (g) × low rate (paise/kg) ÷ 1000
- high = weight (g) × high rate (paise/kg) ÷ 1000
- spoken price = middle of low and high, rounded to the nearest Rs 10

Example: 1.7 kg circuit board at Rs 180–Rs 220/kg → Rs 306–Rs 374 → **"about Rs 340"**.

Photos — a decision we had to make: the plan said to check whether Firebase's photo storage is free. We checked Firebase's official documentation: since **3 February 2026**, free-plan projects cannot use Cloud Storage at all. So we used the plan's backup method:

- the full photo (1280 px, compressed) stays **on the phone**;
- its **SHA-256 fingerprint** is taken the moment it is captured (this goes into the tamper-proof receipt);
- a small thumbnail (480 px, under 200 KB) is saved in Firestore, which **queues it offline** and uploads it when internet returns — no extra upload code needed.

Phase 5 — Today's rates board ✓

Shows every material with its Rs /kg range and a **trend arrow**: green ↑ (price went up more than 5% since last week), red ↓ (went down more than 5%), grey → (about the same). Tapping a tile says, for example, "Circuit board: 180 to 220 rupees per kilo. Rate went up." The arrow, colour and spoken word always go together, so colour is never the only signal. Works offline from the saved price list.

Phase 6 — Finding the best recycler ✓

Ranking formula (PLAN.md §6.3), for one material:

Part	Weight	How it's measured
Rate	45%	This recycler's rate ÷ the best rate on the list
Distance	35%	1 − (distance in km, capped at 25) ÷ 25
Pickup	20%	1 if they collect from the collector, else 0

- **Unauthorised recyclers are hidden.** The whole point is to move scrap into the formal, safe route.
- Distance is the straight-line distance (haversine formula) — no maps SDK, so it works offline.
- Ties are broken by higher rate, then nearer, then name, so the list never jumps around.
- Each card shows: rank, **authorised** badge, distance, rate per kg, **how much you'll get for this lot**, a big **Call** button and a **Choose** button. Tapping the card reads it all out.

Location: the app first explains out loud why it needs location. The collector can tap **My location** (GPS works without internet) or simply **pick their area** from a list if they don't want to share location.

Phase 7 — Tamper-proof handover with QR, and recycler mode ✓

Collector side (S11 Receipt): when the collector chooses a recycler, the app creates a **handover record** (lot, weight, agreed price, time, GPS, photo fingerprints) and seals it with a **SHA-256 hash**:

`v1 | id | collector | recycler | material | weight | price | time | lat | lng | photos-hash` → SHA-256 → 64-character fingerprint

The screen shows this as a **QR code** plus a short reference like **A3F9-1C2B**, which is also spoken one character at a time so it can be read over the phone. A status card shows **amber clock = waiting** until the recycler confirms, then **green tick = confirmed**, with the amount and how it was paid.

Recycler side (recycler mode):

Screen	What the recycler does
R1 Home	Three tiles: Scan QR, My rates, Recent deals
R2 Scan + verify	Scans the collector's QR. The app **re-computes the hash from the QR's data**: green "Record is genuine", or red "Warning! The record was changed". Then the recycler enters the weight they measured and taps **Cash**, **UPI** or **Later** (money due)
R3 My rates	Changes their Rs /kg per material with −10/−1/+1/+10 buttons

Why it can't be cheated: if anyone changes even one digit (for example the weight from 2 kg to 9 kg) the fingerprint no longer matches and the recycler's phone shows a red warning. This needs **no internet**, because the QR carries the whole record.

Automatic warning flags (rule-based):

Flag	When
Record was changed	The hash doesn't match
Weight mismatch	Measured weight differs from the declared weight by more than 10%
Price far from market	Agreed price is below 0.7 × the board's low price or above 1.3 × its high price

Design change from the plan: the recycler writes a separate **Confirmation** record instead of editing the collector's handover. This works even when **both** phones are offline, and the collector's original record stays frozen (the security rules forbid changing it).

Phase 8 — Earnings ledger ✓

Goal: the collector always knows what they earned and what they are still owed.

- Three big totals at the top: **Got today**, **Due today**, **This week**; plus **Total still due** shown separately in amber.
- The speaker reads: "Today you got 640 rupees, 200 rupees still due. This week 1,100 rupees."
- Every confirmed handover automatically becomes a ledger entry (with an ID fixed to that handover, so it can never be counted twice).
- **"Sold elsewhere"** button: pick what was sold, type the rupees on the big keypad, tap **Cash** or **UPI**. This keeps the record honest when scrap goes to a local dealer.

- "Today" and "this week" use the phone's **local time** (weeks start on Monday), while everything is stored in UTC.

Phase 12a — AI that recognises the scrap from the photo ✓

What it does: after the collector takes a photo, a small AI model **on the phone** (no internet needed) guesses what the scrap is. If it is at least **70% sure**, that picture tile moves to the top with a green border and a badge like **"Suggested 92%"**, and the app says "Looks like Mixed plastic, 92 percent sure. Tap it if it is right." If it is less sure, it suggests nothing. The collector always makes the final choice.

How we trained it (on the team's Mac):

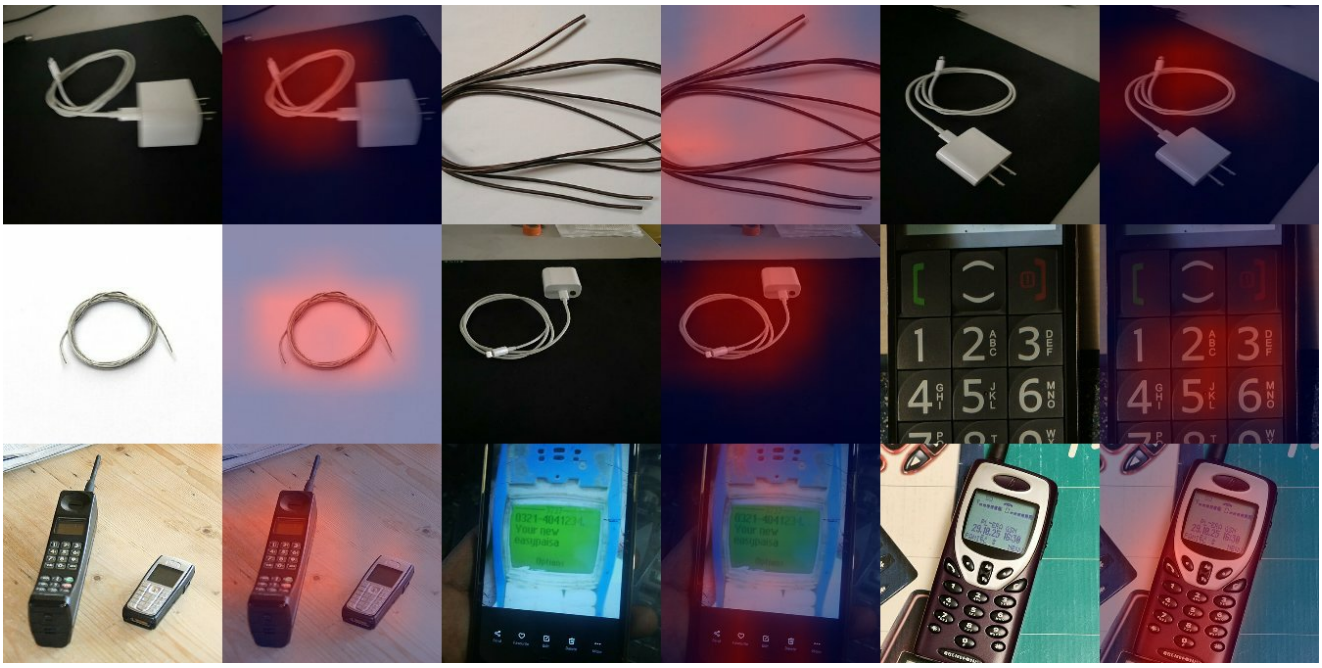
1. **Photos:** 744 photos in 9 folders: the 8 scrap types plus **OTHER** ("not scrap": hands, tables, bags). About 600 are free photos from **Wikimedia Commons** (every author and licence recorded in `data/photos/CREDITS.csv`) and the rest were taken by our team. **We looked at every downloaded photo** and removed about 300 unsuitable ones (people, charts, roofs, power lines).
2. **Model:** MobileNetV2, a small image model made for phones, already trained on millions of general photos. We only taught it our 9 categories ("transfer learning").
3. **Close-up practice:** during training it also sees random **zoomed-in crops**, so it can recognise a phone from just its **camera, ports or buttons**, or a cable from just its end.
4. **Honest test:** 20% of the photos were **held back and never used for training**. The final phone file was tested on them.
5. **Size:** the finished model is **2.7 MB** (int8 TensorFlow Lite).

Result: 80% correct on 137 photos it had never seen.

Category	Accuracy	Category	Accuracy
Phone battery (Li-ion)	100%	Circuit board	75%
OTHER (not scrap)	100%	CRT TV	73%
Copper wire / cable	86%	Vehicle battery (lead-acid)	57%
Mixed plastic	81%	Motor	50%
Phone	79%		

Weak spots, honestly: lead-acid batteries (only 37 photos) get confused with lithium batteries, and motors (42 photos) are hard. More team photos will fix this. Several plastic photos show the same bottle, so the plastic score may be higher than real-world use.

What the AI looks at ("heat maps"): to prove it isn't guessing from the background, we coloured the area of each photo that made the decision. Red = what the AI looked at. It focuses on the **cable coil, the phone keypad and screen**, not the table.



Heat maps: left = photo, right = red where the AI looked. Cables (top), phones (bottom).

Real data instead of made-up data

Real authorised recyclers. We replaced the fake recyclers with the **8 authorised e-waste recyclers of Uttarakhand** from the official **CPCB list (as on 08-06-2023)**, for example M/s Attero Recycling (Bhagwanpur, Roorkee), M/s Root Recycling (Padli Gurjar, Roorkee), M/s Starto Metal Recycle Plant (Roorkee) and M/s Resource E-Waste Solution (Bhadrabad, Haridwar). Names, addresses and registration details are exactly as in the government list. The list gives no phone numbers or rates, so **we did not make any up**: the app shows "Rate not known" and the call button stays off until a verified number is added. Map positions are approximate (village or town centre, from OpenStreetMap).

Because the real recyclers publish no rates, we changed the ranking rule: **authorised recyclers with unknown rates are now shown too**, ranked by distance and pickup only. Unauthorised ones are still always hidden.

Real scrap prices. 7 of 8 prices now come from published scrap-rate websites (2025–2026). Each row in `data/prices.csv` records where the number came from:

Material	Rs per kg	Source (simplified)
Circuit board	80 – 200	recyclinginindia.com, May 2026
Copper wire / cable	100 – 450	reuze.in (wiring) and recyclinginindia.com (copper with insulation)
Phone battery	80 – 130	reuze.in (battery)
Vehicle battery	60 – 120	todaypricerates.com (lead-acid)
CRT TV	3 – 10	Rs 50–200 per TV ÷ about 20 kg
Mixed plastic	8 – 14	recyclinginindia.com; a Roorkee dealer lists Rs 13
Old phone	125 – 310	Rs 20–50 per dead phone ÷ about 0.16 kg
Motor	40 – 55	**Placeholder**: no published rate found

Real facts for the presentation (sources in `data/regional_facts.md`):

Fact	Figure
E-waste generated in India, 2024–25	**13.97 lakh tonnes** (CPCB, Parliament reply)
Uttarakhand e-waste collected/processed, 2024–25	**1,13,562 tonnes** (Parliament reply)
Authorised recyclers in Uttarakhand	**8**, total capacity **1,53,068 tonnes/year** (CPCB list)
Share of recycling done by the informal sector	**about 90–95%** (ICRIER study; formal share is rising)

Features added after testing on a real phone

We installed the app on a **Motorola Edge 50 Fusion (Android 16)** and tested it live. Based on what we saw, we added:

Feature	What it does
Logout	A button on the home screen. It asks "Do you want to log out?" with big Yes / No buttons (spoken). If some data has not been sent yet, it refuses and says "turn on internet first", because logging out then would lose that data.
Back stops the voice	Pressing Back (button or swipe) stops any speech immediately; the app then waits quietly for the next tap.
Voice on / off switch	A switch on the home screen, remembered on the phone. When off, the app does not talk by itself, but the speaker button on each screen still reads that screen out when tapped, so a person who can't read can always get help.
Type the weight in kg or grams	Besides the +/− buttons, a keypad button lets the collector type "350 grams" or "12 kilo". Below 1 kg the app shows and speaks grams.
Live price while setting weight	The fair price ("~Rs 300") is shown under the weight and on the Next button, and changes as the weight changes.

Problems found on the real phone, and fixes:

Problem	Fix
Login failed: "SMS unable to be sent until this region enabled"	Firebase blocks SMS to every country by default. We allowed **India** in Authentication → Settings → SMS region policy.
Screen titles were white on white (invisible)	The title style had no colour set; we gave it the theme's text colour.
The Android build failed after adding the AI library	The library mixed Java 11 and Kotlin 17 settings; we told the build tool to accept that.
A row on the weight screen didn't fit a small 360-pixel-wide screen	Caught by an automatic test on a phone-sized screen; we simplified the row.

7. Testing

Every important piece of logic has automatic tests that run with `flutter test`. Current status: **109 tests, all passing**, and `flutter analyze` reports no issues.

What is tested	Examples
Theme	Buttons are at least 56 dp
Language files	All 12 languages have exactly the same 123 sentences as Hindi, with the same {slots}; every language name is different

What is tested	Examples
Language screen	First page has exactly 4 choices; More shows all 12; Urdu turns the layout right-to-left
AI recogniser	labels.txt is read correctly; centre crop is exact; suggestion only at 70%+; OTHER is never suggested; several photos are averaged; the suggested tile comes first and is spoken
Real data	Real recyclers carry the CPCB reference and no invented phone or rates; unknown rates are listed after known ones
Logout	Asks first; No keeps you logged in; Yes forgets PIN and role but keeps the language; refused while data is unsent
Voice	Voice off = nothing automatic but the speaker button still works; switching on says so; Back stops speech
Weight	Typing 350 grams and 12 kilo works; grams shown below 1 kg; live price on the Next button
Language switching	Tapping a language changes the text instantly and remembers the choice
Data models	Every record can be saved and read back unchanged; unknown values fall back safely
Seed data	Prices are whole paise and low ≤ high; missing recycler rates are skipped; every row is marked placeholder
Offline first run	With no database at all, the app still gets the material list
NumberSpeaker	0, 7, 40, 99, 100, 340, 1000, 1340, 99999, 1,50,000 and 1 crore
Voice fallback	Recorded clips win; a missing number clip doesn't play half a sentence; Marathi TTS; Hindi fallback
Onboarding	Fresh install → language (tap twice) → login; PIN survives a restart and only the right PIN unlocks; recycler mode only for invited numbers
Home	4 tiles, each at least 56 dp, instruction spoken on open
Price	1.7 kg × Rs 180–220 = about Rs 340; rounding to Rs 10; no paisa lost; 500 kg lots stay exact
Sell flow	Category → weight (+½) → "about Rs 300" spoken, with no typing at all
Trend	Up above +5%, down below −5%, exactly ±5% counts as flat
Ranking	Unauthorised and missing rates hidden; formula checked; 25 km cap; ties stable; top 5 only
Hash and QR	Same input = same hash; changing **any** field changes it; tampered weight or price is caught; random QR codes rejected
Warning flags	Weight off by more than 10%; price outside 0.7×–1.3× of the board
Recycler screens	Genuine QR shows green; tampered shows red; another recycler's deal cannot be confirmed
Ledger	Paise totals; due shown separately; day starts at local midnight; week starts Monday; no double counting

App size

Honest note: adding the on-phone AI (about 7 MB of TensorFlow Lite code plus the 2.7 MB model) pushed the main APK over the target. Shrinking it is planned for the polish phase.

The plan's target is a release app under **30 MB**, so it downloads quickly on a cheap data pack. We build one APK per phone processor type ("split per ABI"), so each phone downloads only what it needs:

APK	For which phones	Size
`app-armeabi-v7a-release.apk`	Older and very cheap 32-bit phones	22.8 MB
`app-arm64-v8a-release.apk`	Most phones from the last ~7 years	27.1 MB, now **37.2 MB** with the AI library and 12 languages
`app-x86_64-release.apk`	Emulators / Chromebooks	29.6 MB

8. Data honesty

- **Recyclers** are real, from the official CPCB list (2023); no phone numbers or rates were invented. One clearly named "DEMO Recycler (test account - not real)" exists only for the two-phone demo.
- **Prices** are published web rates with a source on every row, except motors (placeholder). They still need checking with 2–3 local dealers (our field study).
- **Materials** list and **training photos**: team choices and free Wikimedia photos with full credits.
- **Translations** were written by the developers and must be checked by native speakers.
- We never invent real business names or phone numbers.
- The QR hash proves a record **was not changed after it was created**. It is not a legal signature, and we say so in the app.

9. Words used in this document

Word	Meaning
APK	The installable file of an Android app
Firestore	Google's online database that also works offline
Offline-first	The app works fully without internet and syncs later
Seed data	A built-in copy of the price/recycler list for the very first run
Hash (SHA-256)	A "fingerprint" of data: change one letter and the fingerprint changes completely
QR code	A square barcode a phone camera can read
TTS	Text-to-speech: the phone reads text aloud
Paise	1/100 of a rupee; we store money in paise to avoid rounding errors
UUID	A random unique ID made on the phone
Security rules	Server-side rules that decide who may read or change each record
Git commit	A saved snapshot of the code we can always go back to
Haversine	A formula for the straight-line distance between two GPS points on the Earth
Ledger	A record of money earned, paid and still owed

Word	Meaning
Thumbnail	A small, low-resolution copy of a photo
AI model	A small program that learned from example photos to recognise objects
Training / held-out photos	Photos used to teach the AI / photos kept aside to test it honestly
Heat map	A picture showing which part of a photo the AI used to decide
CPCB	Central Pollution Control Board, which publishes the list of authorised recyclers

10. What's next

Phase	What it adds
Two-phone test	Collector phone + recycler phone (+91 99999 00002): QR scan, confirm, ledger update after sync
More team photos	OTHER photos of the demo table and floor; more batteries and motors, to raise AI accuracy
Shrink the app	Bring the APK back under 30 MB
9	Safety guide: picture cards with audio (no burning cables, don't break TVs, batteries away from fire/water, gloves for circuit boards)
10	Sync robustness checklist (airplane mode at each step, app killed mid-flow, phone restarted)
11	Polish: check every screen against the design rules; largest font size; release APK size
12	Optional: on-device scrap recogniser, admin price editor, recorded Garhwali audio, earnings comparison chart
13	Demo mode and a 3-minute demo script
Team (field study)	Replace placeholder prices and recyclers with real data; record Garhwali clips; test with a real collector